

N-Queens Constraint Satisfaction Methods

Purpose

The purpose of this project is to service classroom lessons on constraint satisfaction problems and algorithmic techniques such as filtering and ordering. Simulation data on the time and space efficiency of different combinations of algorithmic techniques is gathered and evaluated to conclude the benefits and takeaways of each method, as well as different specific cases that each might be used in. This data is formatted in different visualizations, including data tables, bar graphs, line graphs, and simulation replays, to demonstrate their performance to a student audience that may or may not be familiar with constraint satisfaction problems, certain algorithms, or algorithmic performance.

This project uses an example of the “N-Queens” problem, which describes a way to find placements of Queen chess pieces on a board size of n rows and columns such that there are n Queens placed that are not contesting each other.

Individual Contribution

Each combination of algorithmic techniques was written and simulated by me, and the visualizations, explanations, and conclusions were created by me, with an exception of the Hennessy & Patterson visual on processor and memory performance.

Principal Tools & Languages Used

All algorithms were written in C++ and simulated on a Linux virtual machine.

Testing

Results of each algorithm were verified using a secondary program that checked the output against the original problem’s constraints.

Material Limitations & Dependencies

The results and conclusions about the algorithmic techniques used are strictly limited to the example of the N-Queens constraint satisfaction problem, and may or may not generalize to other constraint satisfaction problems.